Hello, I'm Ellen Terry. I'm a retired Professor of Bayesian Probability and Networks - this is my channel exploring Emergent Systems. Please subscribe, and contact me here, or on the Github. I'm glad you're here.

To level-set, this channel is targeted for graduate STEM, and there will be math. This first session explores Emergent Geometries in Biological Systems. It is not a research release, and the application code is prototype quality... just an exploration, for information, and inspiration. I also want to encourage Universities and students to pursue Interdisciplinary BioPhysics studies.

Let's get started:

In a recent paper titled: 'Universal gravitational waves from interacting and clustered solitons (see references), Lozanov ɛt.al demonstrated how non-linear self-interactions caused the primordial universe to fragment and condense into stable density fields, curving spacetime and seeding gravitational wells where galaxies, stars, and planets formed.

Here is what that looks like when a density field organizes: these images come from the Max Planck Institute. The halos are attractor basins in the gravitational fields (these basins are mega-parsecs across, and will eventually become galaxy clusters). The filamentary streams are mass flows, channeling dark matter from the low-density voids into high-density halo attractors. This field with attractor basins, is an emergent geometry, and that will be a theme throughout this session, and this channel. I wanted you to see it in it's raw, primordial form. Just blows my mind.

These same geometries govern our biosphere and the organisms in it. That's a long way down - 14 billion years and 15 orders of magnitude. But, cellular processes today, do not operate in isolation nor are they completely governed by biochemical and epigenetic controls.

In this session, we'll build a model of the Human Immune System where epigenetic, thermodynamic and electromagnetic dimensions ... are coupled into one dynamical system. Then we'll track state and motion, across different views and scenarios, to see if a consistent story emerges, around the central question: 'Do these fields affect life systems?' And if so, 'Is there a gap in our current understanding that could affect diagnosis and procedure?'

Before we start building models, and since this is a interdisciplinary audience, we need to level-set on the biology. Let's start with the Immune system. A basic... very basic... we're talking 'Muppets level', overview:

At the top of this diagram, you see a Helper T-Cell interacting with a B-Cell. The B-Cell acts as an Antigen-Presenting Cell (APC), and the diagram maps out the path it follows.

To prevent accidental immune responses where the body attacks self-tissue - the system enforces a rigorous three-signal checkpoint process:

Signal 1: Antigen Recognition.

The B cell captures an antigen and displays its peptide fragments on the B-Cell surface. A matching T-Cell binds to these peptides, tripping an alarm.

Signal 2: Co-Stimulation (The Safety Check).

To confirm this is a real infection and not a false alarm, surface receptors on the two cells lock together. This handshake licenses the B cell to enter the germinal center (GC). This is where the cell begins Somatic Hypermutation (SHM) - an optimistic process that randomly mutates the cell's DNA at high speed to find a design that tightens its grip on the antigen.

Signal 3: Cytokine Selection and Fate Assignment.

Once mutation produces an optimized antibody, the T cell floods the interface with specialized cytokines (an immune response protein). The strength of this final cue judges the quality of the new design and assigns the cell's fate to one of two paths:

The Plasma Cell Path (High Affinity): If the antibody binds tightly, intense signaling forces the cell to leave the GC and become an antibody factory, pumping out thousands of antibodies per second. This is the big threshold transition we'll focus on here - modeling the B-Cell path from the GC to Plasma cell.

But, to close this section, the second path the B-Cell can take is the transition to a Memory Cell (Moderate Affinity). A different signaling balance sets the cell aside as a longterm sentinel, waiting in reserve for future infections.

OK, that's good - we just needed a sense of the B-Cell path from GC to Plasma. Now for the physics, which will lead us into the modeling section.

I read the Feynman CalTech lectures, one summer vacation in High School, and I still have that box set - a bit yellow and dog-eared now, but it was an absolute epiphany for me. In Feynman's path-integral formula, a particle explores all possible paths across the action landscape. Then, when Born and Schrödinger gave us the density field $\vert{}\psi\vert{}^2$, they showed that physical reality at that scale is governed by field geometries.

But it was Einstein who first revolutionized our views of space-time, demonstrating that fields are not passive grids. They are Dynamic Geometries... that Actively Shape... the Trajectory of Matter.

Einstein's field equations (top left) give shape to the energy-momentum tensor T, transforming energy Gradients into a physical curvature, or landscape. How does energy and matter move across that landscape? Thats defined by the Geodesic Motion equation (top right).

Moving down a row, we scale the equations all the way down, from the cosmic to the biological cell level. We carry the thermodynamic and electromagnetic fields down with us. Remember: Fields are, by definition... always... everywhere. At this level, we want to track the biology too, so we add epi-genetic and now we have 3 dimensions.

And those dimensions create a Landscape and Path at the Biological level. Think of the next 3 sections as a 'user guide' for the exploration panels we'll be using. First, the Landscape panel here. Then, we'll move down a level to Individual Gene Kinetics (these 2 panels will sit side-by-side in the animations and analysis). Then I'll do a quick review of the underlying equations so you know what's on the Github in case you want to run it yourself, or adapt it to other scenarios (there's also a tech paper on the Github). Finally, we'll put the scenario in motion and disrupt it (gene knockout) to see what happens.

Here, I'm showing how the Landscape and Motion equations are connected to their outputs. The top equation produces the landscape surface, and the motion equation below produces the cell paths - the same path we covered in the 'Immune System Overview'. And below that is the vector y that feeds the equations at this level.

The workhorse function in the Landscape is Pss which solves using a Fokker-Planck equation, and that produces a steady-state 'Density' field that gives us the probability of finding a cell in a specific state. The output of Pss is transformed using a negative log-likelihood, which inverts the density field so that the most likely areas, are darker and at the bottom. These are the 'Attractor Basins' (which is a definitive property of Emergent Systems).

In the motion equation below, the gradient term $-\ D\nabla U(y)$ acts as the conservative drift, pulling the state vector downhill into the attractor basin like a ball settling into a valley. That sets the general descent, but life is a non-equilibrium, entropy production system. To capture that reality, I integrate Wang's non-equilibrium curl flux (J_{curl}) a rotational current that drives eddy-like circulation through the state space, sustained by continuous thermodynamic dissipation.

The y vector feeds it all - all equations, and panels. It solves 7 differential equations simultaneously, and then pro-jects a physical phase space: The vertical Z-axis is the quasi-potential energy elevation, $U(y)$, think of it as the metabolic hill that the cell must climb to move from one phase space to another. The horizontal Epigenetic X axis tracks genetic expression, and the bioelectric axis Y, tracks membrane potential, V-mem. That's the 3-axis Landscape that the B-Cell must navigate to become a Plasma-Cell.

Moving to a continuous-time view, the same core y vector drives this multi-axis physical kinetics panel. Using two interdependent vertical scales here, the Left axis is multipurpose:

First it traces gene expression , which is also consolidated into the Epigenetic Index (X-axis) on the Landscape panel, but here we can see the individual genes.

Second, it traces Mitochondrial Fusion, M-fuse - yellow line. (mitochondria will fuse into networks to increase energy capacity). This has to be in place, in advance of energy demand.

Third, it measures Entropy Production Rate, ΔS, the red dotted line. This is the physical heat generated by ion pumps charging the membrane through the mitochondrial network. This is the thermodynamic cost paid - the unavoidable dissipative burn - as the cell navigates the landscape.

Now, the Right Axis. This traces membrane potential (V-mem) along the dashed blue line, which maps directly to the y axis on our landscape. As the B cell hyper-polarizes to -80 mV, it maximizes the inward electrochemical driving force for calcium entry - creating a localized information-density field, establishing the geometry of the landscape.

Finally, here's the underlying system of equations in the y vector that creates the panels we just reviewed, so we're repeating the same components and dynamics as we step down through the levels - you should start to notice repetitive themes.

The y vector begins with the Martinez Genetic Regulatory Network (GRN) models, using the Hill transforms (foundational immunology stuff). Then I add thermodynamic and electromagnetic equations to the system. I'm not going to dig into every equation here, I just want to highlight a few roles, functions and integration points to watch. Then, we'll put it all together and run simulations.

The first two are Pax5 and BCL6. Note that they have numerator relationships so they work together. Their role is preserving the GC processes, because SHM is rapidly mutating DNA (recall from Immune System overview) and there are other systems on a mission to remove cells with mutated DNA. So, the B-Cell needs protection. I call this the 'GC' couple

The next coupling starts with IRF4, which is triggered by Signal 3 (also recall that last signal from the overview, and note S-3 in the equation). That brings in Blimp1, the 'Secretory Master'. which has IRF4 in its numerator - So, these two are the 'Plasma' couple. Once S-3 is fired, the 'Plasma Transition' couple turns on, driving the 'GC' team down."

So then, the cell is committed to plasma, which means massive antibody secretion, which demands energy... a Lot of Energy. so the mitochondrial network will have to expand - and when scaled to thousands of cells, this requires coordination and communication of information.

In the IRF4 equation, there's a term Phi-NFat, which is a calcium-sensitive transcription factor, activated when the cell hyper-polarizes - as discussed in the previous kinetics panel section, on the V-mem axis. And as noted there, this connects the localized density-information field. This Phi-NFat threshold motion changes the landscape, but it does not define direction for the protein, the gene, the network, or the landscape... and it is not causal. Direction is defined by the geometry and that is the causal map.

The bottom equation is outside the y vector. Entropy production (ΔS), is a monitoring metric. It tells us if the overall system is doing what it's supposed to do... produce entropy.

OK, that's the equations. Let's watch this in motion.

We are going to watch two simulations: first, a normal B-Cell process, for baseline; then, a targeted CRISPR knockout of Pax5 to expose what happens when GC shielding collapses. I'll try to keep us focused on the GC and Plasma couples for epigenetic motion. There are additional genes being traced for deep dives and questions, but best to start with those two couples.

Here are four things to watch for on the Left Panel:

1. Watch how the GC and Plasma couples cross at Signal 3.

2. The M-fuse yellow line creates the substrate that will support Delta S energy-entropy (red dotted line). If the mitochondrial network is sufficient it will not need to expand, and it generally starts that way until energy demand increases Also watch how Delta-S moves slightly ahead of genetic transitions (a causality cue).

3. The dashed blue, membrane potential, V-mem, is negatively charged. So the line will move down as membranes hyperpolarize. Watch how it also moves slightly ahead of genetic transitions (causality cue again).

4. Over on the Right Panel (Landscape): Watch how the cell navigates up and over the ridge, to the attractor basin. It's a smooth path in the normal process, but gets more interesting when we knockout Pax5 in the second scenario.

Let's walk through it:

The GC State

When the B-Cell is in the Germinal Center, traces are mostly flat. The GC couple (red and orange), is at high expression level (SHM is working), and the Plasma couple (blue and green) expression is low. V-mem (blue-dash), is depolarized at $-35 mV.

Around 9 seconds, the V-mem, and the Delta S (red dot)... begin to move (M-fuse yellow will catch up later - mitochondrial network config is sufficient for now). These are the only traces in motion. Over on the landscape, the cell is in the Germinal Center well, but is beginning to climb up the ridge.

Signal 3 Arrival

At t = 10, Signal 3 arrives. Now V-mem is polarizing fast (the density-information field is connecting to the gradients). Then, the cyan trace (IRF4) ignites and moves up. The red dotted entropy line (Delta S) surges. Mitochondrial networks, M-fuse yellow line, begin expanding to build metabolic manifolds for the transition and antibody production ahead. On the landscape, the cell is clearing the ridge without stalling.

Terminal Descent and Attractor Lock

The GC couple is 'Suppressed' now, while the Plasma couple is fully 'Expressed'. The M-fuse yellow line holds steady to support continued antibody secretion. On the landscape, the cell settles into the Plasma Attractor Basin.

OK, that's a normal process. Remember, there are thousands of B-Cells doing this: Both as individual actors (the redundancy of evolution), and as informed agents of the whole (the collective system). Now, let's disrupt this and see what happens:

GC State

Same as the first: The GC couple are in the Well, and V-Mem rests at -35 mV.

Lineage Shield Collapse and Premature Ignition

At t = 10, I knockout Pax5 (virtual CRISPR), so the GC team collapses. Notice that entropy production, red dotted Delta S, shoots right up the Signal 3 line (the lines are close, so a little hard to see - but you should be able to see it shoot up to 4, then drop to 3, and then recover - moving into it's normal path).

Despite the disruption, V-mem hyper-polarizes, connecting to the information-density field. It will sometimes become a little erratic, depending on the endogenous state, but it will recover quickly. Mitochondrial fusion increases as before. Both paths are slightly accelerated. It reminds me of when I'm driving and take a wrong turn... 'proceed to route', 'proceed to route'.

I would describe this as disruption, short confusion, then recovery and rerouting. Some of it is biochemical homeostasis, but I don't buy that as the root cause - I have established, across every perspective, that: it is the geometry that sets direction, and the geometry is defined by all three dimensions here.

Reviewing: Cell fate transitions are not purely genetic decisions. The landscape is created by field and motion geometries - mechanisms that have driven the universe for fourteen billion years. Life is a continuation of these physical laws, not an exception to them.

Here, the B-cell trajectory is shown once more, contrasted against the classic gene regulatory network path for the exact same transition, plotted in red. Look at the divergence. Those gaps are not noise - they are unmodeled physics, and that is where our engineering opportunities lie.

Whether your focus is engineered cell therapies, biomanufacturing, drug discovery, tissue engineering, or direct clinical intervention... adopting emergent dynamical models reveals process improvements.

Those gaps exist because conventional workflows assume thermodynamic energy is free and treat ion channels as passive readouts. The clinical result of that blind spot is familiar across the industry: premature cell exhaustion, transcriptional arrest, and phenotypic drift. The cell stalls on the slope or veers off-target simply because its physical dimensions were never aligned with its genetic program.

We opened this session fourteen billion years ago in the early universe, long before the assembly of galaxies or planetary systems. Space is not empty; it is populated by coupled physical fields that channel energy through geometric densities into dissipative structures - both

physical and living. That is our origin, and that is the thermodynamic directive of life. It is an extraordinary reality that requires no embellishment.